CLASS 10 — MATHS · FORMULA SHEET

Chapter 8: **Introduction to Trigonometry** | Quick revision, one page

Trigonometric ratios (right angle at B, angle A)

- $\sin A = \dfrac{\text{perpendicular}}{\text{hypotenuse}},\ \cos A = \dfrac{\text{base}}{\text{hypotenuse}},\ \tan A = \dfrac{\text{perpendicular}}{\text{base}}.$
- $\operatorname{cosec} A = \dfrac{1}{\sin A},\ \sec A = \dfrac{1}{\cos A},\ \cot A = \dfrac{1}{\tan A}.$
- $\tan A = \dfrac{\sin A}{\cos A},\ \cot A = \dfrac{\cos A}{\sin A}.$
- $0 \le \sin A \le 1,\ 0 \le \cos A \le 1;\ \sec A, \operatorname{cosec} A \ge 1.$

Values of standard angles

- sin: $0,\ \frac{1}{2},\ \frac{1}{\sqrt{2}},\ \frac{\sqrt{3}}{2},\ 1$ for $0^\circ, 30^\circ, 45^\circ, 60^\circ, 90^\circ$.
- cos: $1,\ \frac{\sqrt{3}}{2},\ \frac{1}{\sqrt{2}},\ \frac{1}{2},\ 0$ for the same angles.
- tan: $0,\ \frac{1}{\sqrt{3}},\ 1,\ \sqrt{3},\ \text{not defined}$ for the same angles.

Complementary angles

- $\sin(90^\circ - A) = \cos A,\ \cos(90^\circ - A) = \sin A.$
- $\tan(90^\circ - A) = \cot A,\ \cot(90^\circ - A) = \tan A.$
- $\sec(90^\circ - A) = \operatorname{cosec} A,\ \operatorname{cosec}(90^\circ - A) = \sec A.$

Identities

- $\sin^2 A + \cos^2 A = 1.$
- $1 + \tan^2 A = \sec^2 A.$
- $1 + \cot^2 A = \operatorname{cosec}^2 A.$

Handy results

- Pythagorean triples $(3, 4, 5), (5, 12, 13), (8, 15, 17), (7, 24, 25)$ give exact ratios.
- To find all ratios from one: draw the right triangle, find the third side by Pythagoras.

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